



Lab: Evaluate Existing EDA of the Companies

Estimated time: 60 minutes

Introduction

FashionMart and **TrendyThreads** have different data architectures to match their business models that is physical retail and virtual e-commerce. Throughout this lab, you will critically analyze their current data environments, including data sources, storage processes, and integration flows.

Critical analysis components include data governance, security, and compliance practices to measure overall system resiliency. With the use of tools such as **Lucidchart**, you will map current architectures and outline inefficiencies, data silos, and integration issues.

The aim is to create best practices and suggest enhancements for a more integrated, scalable, and efficient enterprise data strategy.

References

Right-click and select **Open link in new tab** to review the enterprise data architecture documents for [TrendyThreads](#), and [FashionMart](#).

This will help you:

- Gain insights into how each company structures, stores, and manages its data
- Understand the key differences between digital-first and brick-and-mortar data architectures
- Analyze the integration challenges and opportunities in merging two distinct data ecosystems

- Identify best practices for designing a scalable and efficient enterprise data strategy

Tools:

You can complete your tasks using any text editor, such as Word, Google Docs, or Notepad.

- If you install Microsoft Word, you can proceed directly with the instructions. However, if you don't have Microsoft Word, you can sign up for a free version of Microsoft 365 online. Follow [these instructions](#) to create your account.
- You can also use Word Online with a Microsoft 365 subscription or access through a free account. If you're new to Word and would like to learn how to get started, you can find helpful resources here: [Getting Started with Word Online](#).
- You may use [Lucidchart](#) or any other diagramming tool for this lab. The free version of Lucidchart is sufficient. You will be responsible for any associated costs if you purchase a paid plan for Lucidchart or any other tool. Please follow the [instructions](#) if you have not yet signed up for a free version of the Lucidchart account.

Recommendation: Before proceeding with this lab, it is important that you finish the lab [Create an EDA Blueprint](#) first, as it provides a solid understanding of how to design an efficient EDA blueprint.

Tasks

Part 1: Documenting the Enterprise Data Architecture (EDA) blueprint for FashionMart

Objective

Create a comprehensive EDA blueprint for FashionMart, capturing key architectural layers, tools, and processes. The blueprint should illustrate the end-to-end data flow, ensuring a clear understanding of data movement, transformation, and security.

Task 1.1: Identify and define key architectural layers

- **Data sources:** Document all sources of data, including POS systems, customer transactions, supplier databases, and third-party data feeds
- **Data ingestion:** Define the tools and processes used to extract, transform, and load (ETL/ELT) data from various sources into storage

- **Data storage:** Specify storage solutions such as on-premises databases, cloud storage, or data lakes
- **Data processing:** Outline how raw data is processed, cleaned, and transformed for analytics using ETL pipelines or data processing frameworks
- **Analytics:** Define the reporting and business intelligence tools used for insights generation
- **Security:** Document data security protocols, encryption methods, and access control mechanisms

Task 1.2: Visualizing the EDA blueprint

- Create a structured blueprint diagram showcasing the flow of data across the layers using Lucidchart or any other diagramming tool of your choice.
- Save the blueprint diagram as **FashionMartEDABluePrint.jpg** for reference in the final project submission

Part 2: Documenting the Enterprise Data Architecture (EDA) blueprint for TrendyThreads

Objective

Create a detailed EDA blueprint for TrendyThreads, incorporating data layers, tools, and processes. This blueprint will help visualize data movement, analytics capabilities, and governance strategies.

Task 2.1: Define key architectural layers

- **Data sources:** Identify data sources, including website interactions, customer purchase history, and social media engagements
- **Data ingestion:** Specify ingestion tools and methods for collecting real-time and batch data from various digital touchpoints
- **Data storage:** Document storage infrastructure such as cloud databases, NoSQL stores, or data warehouses
- **Data processing:** Define data transformation methods, including ETL workflows, real-time streaming, and batch processing
- **Analytics:** Detail the business intelligence and analytics platforms for insights and performance tracking
- **Personalization & engagement:** Outline how AI-driven models enhance customer personalization and experience

- **Security:** Describe security frameworks, including encryption, authentication, and access management
- **Governance:** Explain data governance policies, compliance standards, and audit mechanisms

Task 2.2: Visualizing the EDA blueprint

- Develop a structured visual representation of the data flow across all architectural layers using Lucidchart or any other diagramming tool of your choice.
- Save the blueprint diagram as **TrendyThreadsEDABluePrint.jpg** for reference in the final project submission

Part 3: Evaluating FashionMart's data architecture: Strengths and areas for improvement

Objective

Assess FashionMart's EDA by identifying its strengths and areas for improvement across key architectural components.

Task 3.1: Analyze key architectural components

- **Data silos:** Identify whether isolated data pockets exist and their impact on decision-making
- **Legacy systems:** Assess the impact of outdated systems on scalability and performance
- **Scalable data architecture:** Evaluate the system's ability to handle growing data volumes
- **Data integration capabilities:** Determine how well different systems and data sources are integrated
- **Robust data governance:** Analyze the policies, compliance frameworks, and governance structure
- **Performance bottlenecks in data processing:** Identify inefficiencies that slow down data pipelines
- **High availability and disaster recovery plans:** Review system reliability, failover mechanisms, and backup strategies
- **Analytics and BI tools:** Assess the effectiveness of reporting and analytics platforms

- **Data security measures:** Examine data encryption, access controls, and compliance adherence

Task 3.2: Document strengths and areas for improvement

- Out of the 9 key architectural components mentioned above, identify the **strengths** and **areas for improvement** for 3 components in FashionMart's data architecture
- If you are using Microsoft Word, right-click and select **Open link in new tab** to download the [template](#) document, and use it to structure your findings. You can follow the same steps to download and use the [PDF format](#) for any other word-processing software.
- Save the assessment document as **FashionMart Strengths and Areas for Improvement.pdf** for reference in the final project submission.

Below are examples of some sample strengths and areas for improvement.

Example of strengths:

Strength	Description	Impact
Scalable data architecture	The current architecture supports seamless scalability.	Supports rapid business expansion.
Data integration capabilities	The current architecture has strong integration between various data sources and systems, providing a single view of business operations.	Supports better decision-making and data consistency.

Example areas for improvement:

Area for improvement	Description	Impact
Data silos	Some business units operate with their own isolated data systems.	• Does not provide a single view of the enterprise. • Reporting and analytics are challenging.

Legacy systems	Older systems are not fully integrated with the new architecture. This is leading to data discrepancies.	• Impacts the data flow across systems. • Limits scalability.
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Part 4: Evaluating TrendyThreads' data architecture: Strengths and areas for improvement

Objective

Conduct a detailed assessment of TrendyThreads' enterprise data architecture by analyzing its core components. Identify strengths and areas for improvement to optimize performance, scalability, and security.

Task 4.1: Analyze key architectural components

Examine the following components of TrendyThreads' data architecture:

- **Data silos:** Assess whether data is fragmented across different departments and its impact on data accessibility
- **Legacy systems:** Identify outdated systems and evaluate their effect on operational efficiency and integration
- **Scalable data architecture:** Determine how well the architecture supports business growth and increasing data volumes
- **Data integration capabilities:** Evaluate the effectiveness of data pipelines, API connectivity, and cross-system integration
- **Robust data governance:** Review compliance with data governance policies, access controls, and data quality standards
- **Performance bottlenecks in data processing:** Identify slowdowns in ETL processes, query performance, and real-time data analytics
- **High availability and disaster recovery plans:** Analyze system resilience, failover mechanisms, and backup strategies
- **Analytics and BI Tools:** Assess the ability of analytics tools to provide meaningful insights for decision-making
- **Data security measures:** Examine security protocols, encryption techniques, and compliance with industry regulations

Task 4.2: Document strengths and areas for improvement

- Out of the 9 key architectural components mentioned above, identify the **strengths** and **areas for improvement** for 3 components in TrendyThreads' data architecture.
- If you are using Microsoft Word, right-click and select **Open link in new tab** to download the [template](#) document, and use it to structure your findings. You can follow the same steps to download and use the [PDF format](#) for any other word-processing software.
- Save the assessment document as **TrendyThreads Strengths and Areas for Improvement.pdf** for reference in the final project submission.

Summary

Congratulations on completing this Lab: Evaluating Existing EDA of the companies. In this lab, you've designed comprehensive EDA blueprints that map out key architectural layers, including data sources, ingestion, storage, processing, analytics, and security for FashionMart and TrendyThreads.

Additionally, you've evaluated the strengths and areas for improvement in each company's architecture, identifying critical opportunities for enhancement.